

## **Basic Information**

**Experiment No.:** 07

**Date:** 24-09-2026

**Title:** Java ArrayList and Its Methods

---

## **Objective / Aim**

To implement "ArrayList" in Java and demonstrate common operations such as adding, accessing, updating, removing, searching, sorting, and clearing elements.

---

## **System Requirements**

### Hardware

- Computer/Laptop
- Minimum 4 GB RAM

### Software

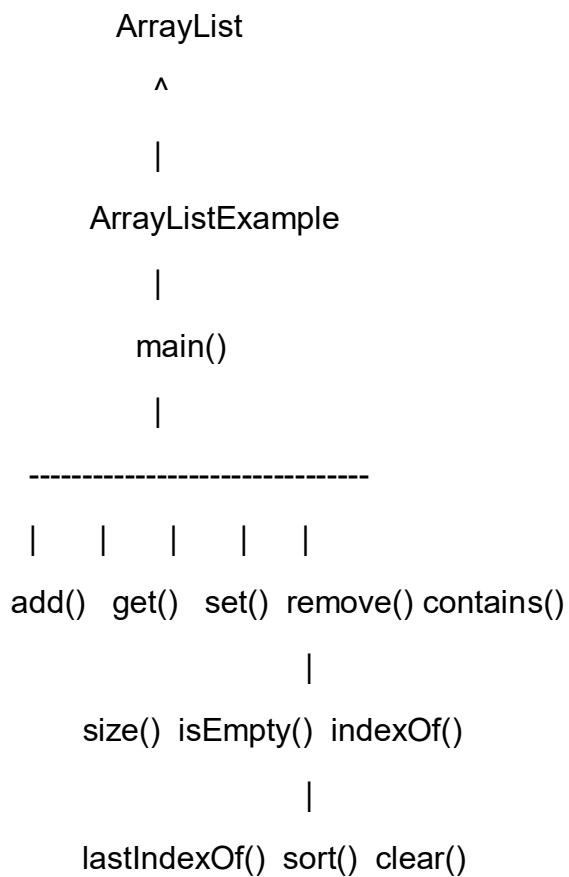
- Java JDK
  - Any Java IDE or Command Prompt
  - Operating System: Windows/Linux/macOS
- 

## **Algorithm / Logic Description**

1. Create an "ArrayList" of strings.
2. Add elements using "add()".
3. Insert an element at a specific index using "add(index, element)".
4. Access and update elements using "get()" and "set()".
5. Remove elements using "remove()".

6. Check elements using "contains()".
  7. Find size and positions using "size()", "indexOf()", and "lastIndexOf()".
  8. Check whether the list is empty using "isEmpty()".
  9. Sort the list using "sort()".
  10. Remove all elements using "clear()".
- 

## UML Class Diagram



## Source Code

```
package collectionframework;
```

```
import java.util.ArrayList;
import java.util.Comparator;

public class ArrayListExample {
    public static void main(String[] args) {

        ArrayList<String> list = new ArrayList<>();

        // add()
        list.add("Apple");
        list.add("Banana");
        list.add("Mango");
        list.add("Apple");

        System.out.println("Original List: " + list);

        // add(index, element)
        list.add(1, "Orange");
        System.out.println("After add: " + list);

        // get()
        System.out.println("Element at index 2: " + list.get(2));

        // set()
        list.set(2, "Grapes");
        System.out.println("After set: " + list);

        // remove(index)
```

```
list.remove(1);
System.out.println("After remove index: " + list);

// remove(Object)
list.remove("Apple");
System.out.println("After remove object: " + list);

// contains()
System.out.println("Contains Mango: " + list.contains("Mango"));

// size()
System.out.println("Size: " + list.size());

// isEmpty()
System.out.println("Is Empty: " + list.isEmpty());

// indexOf()
System.out.println("Index of Mango: " + list.indexOf("Mango"));

// lastIndexOf()
System.out.println("Last Index of Mango: " +
    list.lastIndexOf("Mango"));

// sort()
list.sort(Comparator.naturalOrder());
System.out.println("Sorted List: " + list);

// clear()
list.clear();
```

```
        System.out.println("After clear: " + list);
    }
}
```

---

## **Compilation & Execution Commands**

Save the file as:

ArrayListExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac ArrayListExample.java

Run:

java collectionframework.ArrayListExample

---

## **Output Screen**

Original List: [Apple, Banana, Mango, Apple]

After add: [Apple, Orange, Banana, Mango, Apple]

Element at index 2: Banana

After set: [Apple, Orange, Grapes, Mango, Apple]

After remove index: [Apple, Banana, Mango, Apple]

After remove object: [Banana, Mango, Apple]

Contains Mango: true

Size: 3

Is Empty: false

Index of Mango: 1

Last Index of Mango: 1

Sorted List: [Apple, Banana, Mango]

After clear: []

---

## **Result**

The "ArrayList" class was successfully implemented, and operations such as "add()", "get()", "set()", "remove()", "contains()", "size()", "indexOf()", "lastIndexOf()", "sort()", "isEmpty()", and "clear()" were successfully demonstrated.